SECTION 1: Identification of the substance/mixture and of the company/ undertaking

1.1. Product identifier

Sorbotech LG 95; 95 AW; 85; R

Product name: Activated carbon
Chemical name: Carbon [CAS: 7440-44-0; EC: 231-153-3]
Registration number: 01-2119488894-16-0042

1.2. Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: Used as a filter material.
Uses advised against: Not determined.

1.3. Details of the supplier of the safety data sheet

Supplier: ACES S.A.
Address: Poland; PL 81-537 Gdynia; Łużycka 10
Telephone/Fax: +48 58 622 63 50 / +48 58 623 30 17
E-mail address of the person responsible for the SDS: reach@aces.pl

1.4. Emergency telephone number

112 (emergency call), 998 (fire brigade), 999 (ambulance)

SECTION 2: Hazards identification

2.1. Classification of the substance or mixture

Classification according to Regulation 1272/2008/EC:
Hazards to man: Does not meet the criteria of classification.
Hazards to environment: Does not meet the criteria of classification as dangerous for the environment.
Hazard from physical and chemical properties: Does not meet the criteria of classification.

2.2. Label elements

Label accordance with Regulation 1272/2008/EC (CLP)
Hazard pictograms, signal words: None.
Hazard statements: None.
Precautionary statements: None.
The names of hazardous ingredients on the label: None.

2.3. Other hazards

The product does not meet the criteria for PBT or vPvB in accordance with Annex XIII of Regulation 1907/2006 (REACH).

May form explosible dust-air mixture if dispersed.

Depending on kind of proceeding and usage (e.g. grinding, transport), in the air can arise "mineral dust". In this case applied material should be treated as dangerous substance.

Wet activated carbon absorbs oxygen from air, which may pose a risk to people working inside tanks with an activated carbon and confined or enclosed spaces without ventilation. Before entering into these places, observe the rules of work in areas with low oxygen,

ensure a relatively efficient ventilation or use self-contained respirators with an independent air circulation.
Contact with strong oxidizing e.g. ozone, liquid oxygen, chloride, permanganate, etc., may cause ignition.

SECTION 3: Composition/information on ingredients

3.1. Substances

Substance name:	**Carbon**
Concentration [%]:	90
CAS Number:	7440-44-0
EC Number:	231-153-3
Index Number:	-
Classification 1272/2008/EC:	-

The product does not contain impurities and additives classified as hazardous in accordance to Regulation 1272/2008/EC.

3.2. Mixtures

Not applicable.

SECTION 4: First aid measures

4.1. Description of first aid measures

Inhalation: Move the affected person to fresh air and keep rested. Seek medical advice if necessary.
Skin contact: Take off contaminated clothing. Wash contaminated skin with water with soap. Seek medical advice if necessary.
Eye contact: Remove contact lenses, if easy to do. Immediately flush eyes with plenty of water, occasionally lifting the upper and lower eyelids. Continue to rinse for at least 15 minutes. Seek medical advice if necessary.
Ingestion: Rinse mouth and give plenty of water to drink. Seek medical advice if necessary.

4.2. Most important symptoms and effects, both acute and delayed

Inhalation: Dust may cause respiratory tract irritation.
Skin contact: May cause skin irritation.
Eye contact: In high concentration product cause serious conjunctivitis and inflammation of the cornea of the eye.
Ingestion: May cause mucous membranes of mouth and throat irritation.

4.3. Indication of any immediate medical attention and special treatment needed

Remove affected person from the environment contaminated by the product. In the event of health problems, consult your doctor or the center of toxicological concern. Provide the information contained in the SDS. If unconscious, do not give anything by mouth.

SECTION 5: Firefighting measures

5.1. Extinguishing media

Suitable extinguishing media: Foam, dry chemicals, carbon dioxide (CO_2), water spray.
Unsuitable extinguishing media: Water jet.

5.2. Special hazards arising from the substance or mixture
Heated to a temperature above 600°C decomposes to carbon monoxide (CO) and carbon dioxide (CO_2). During combustion irritation and hazardous products may be formed, e.g.: carbon oxide. Avoid inhalation of combustion products, because they may pose a health risk. Wet activated carbon absorb oxygen from air, thus is possibility of dangerously a low oxygen levels.

5.3. Advice for firefighters
Wear full protective equipment and self-contained breathing apparatus with independent air circulation. Containers exposed to fire or high temperature cool with water and if possible remove from the danger zone. Wet activated carbon absorb oxygen from air, thus is possibility of dangerously a low oxygen levels. Protect drains, surface waters and soil from pollution.

SECTION 6: Accidental release measures
6.1. Personal precautions, protective equipment and emergency procedures
For non-emergency personnel: Access of non-emergency personnel to the area of accident should be restricted until the completion of the disposal of the product. Wear appropriate personal protective equipment. Avoid dust formation.
For emergency responders: Wear appropriate personal protective equipment. Prevent formation and inhalation of dust. Provide adequate ventilation.

6.2. Environmental precautions
Secure the gullies. Prevent contamination of surface water and ground. In the event of any serious pollution of the environment, notify the appropriate administrative authority, control and rescue services. Used containers should be disposed by delivering to eligible organizations.

6.3. Methods and material for containment and cleaning up
Secure the damaged packaging. Collect mechanical and dilute with plenty of water. Collected product put in a substitute container and direct to the destruction. Provide well-ventilation in contaminated area.

6.4. Reference to other sections
Disposal - see Section 13. Personal protective equipment - see Section 8.

SECTION 7: Handling and storage
7.1. Precautions for safe handling
Do not eat, drink, smoke or take drugs at work. Wash your hands before break and after working with the product. Wear appropriate personal protective equipment. Take off contaminated clothing and wash it before reuse. Provide adequate ventilation, general and local, during working with the product. Do not pulverize on a naked flame or incandescent material. Keep away from ignition sources – no smoking. Install filters in a ventilation system, before released into an atmosphere.

7.2. Conditions for safe storage, including any incompatibilities
Keep in properly labeled, factory sealed containers, with a label which complies with current regulations. Store in a dry magazine. Storage temperature must not exceed 80°C. Limit exposure to air and moisture to protect against decay. Restrict access to storage areas for wet activated carbon. In areas where it is stored wet activated carbon is recommended to

control the level of oxygen. Bulk storage should take place in specially designed silos. Keep out of reach of children. Avoid contact with strong acids and strong oxidizing agents.

7.3. Specific end use(s)
Used as a filter material.

SECTION 8: Exposure controls/personal protection
8.1. Control parameters

Substance name	TWA	STEL	BLV
Dust	10 mg/m3 (inhalable dust) 4 mg/m3 (respirable dust)	-	-

Legal basis: Ordinance on maximum permissible concentration and intensity of harmful factors in the work environment in accordance with national limit values.
EH40/2005 Workplace exposure limits, second edition, published 2011, ISBN 978 0 7176 6446 7.

Monitoring procedures: Use methods described in European Standards.

8.2. Exposure controls
Mandatory general regulations on occupational health. Do not allow to exceed the normative concentrations of hazardous constituents in the workplace. After work, wash and clean the surface of the body and clothing. Do not eat, drink, smoke or take drugs at work. Take off immediately all contaminated clothing and wash it before reuse. Wash hands and face before break and after working with the product. Provide adequate ventilation, during working with the product.

Eye/face protection: Wear suitable tight protective goggles with face shields (in accordance with EN 166).
Skin Protection: Wear suitable protective gloves made of nitrile rubber (in accordance with EN 374). Use protection cream.
Respiratory protection: Wear appropriate respiratory protection when working under a dusty atmosphere: mask or respirator with an air filter P1 type.
Thermal Hazards: Usually not required.

Used personal protective equipment should meet the requirements of local/regional/national laws. The employer must provide personal protective equipment appropriate to the type of work and meeting all requirements, including maintenance and cleaning.

Concentrations of hazardous substances in the workplace should be monitored in accordance with acknowledged test methods. Mode, method, type and frequency of testing and measurement of harmful factors in the working environment should meet the requirements of local/regional/national laws.

Environmental exposure controls: The large amount of product should not be allowed to penetrate through the ground water, sewage, waste water or soil.

SECTION 9: Physical and chemical properties

9.1. Information on basic physical and chemical properties

Appearance:	Black, dusty granules
Odour:	Odourless
Odour threshold:	Not specified
pH:	Not specified
Melting point/freezing point:	>1000°C
Initial boiling point and boiling range:	>1000°C
Flash point:	Not specified
Evaporation rate:	Not specified
Flammability (solid, gas):	Not specified
Upper/lower flammability or explosive limits:	Not specified
Vapour pressure:	Not specified
Vapour density:	Not specified
Relative density:	Density: 2.31 g/cm^3
Solubility:	Insoluble in water
Partition coefficient: n-octanol/water:	Not specified
Auto-ignition temperature:	350-450°C
Decomposition temperature:	Not specified
Viscosity:	Not specified
Explosive properties:	May form explosible dust-air mixture if dispersed
Oxidising properties:	Not specified

9.2. Other information

Bulk density: 350-550 kg/m^3

SECTION 10: Stability and reactivity

10.1. Reactivity

Reacts with strong acids and strong oxidizing agent.

10.2. Chemical stability

Stable under normal conditions of storage.

10.3. Possibility of hazardous reactions

Contact with strong oxidizing agents may cause rapid ignition/explosion. May form explosible dust-air mixture if dispersed.

10.4. Conditions to avoid

Do not expose to air at temperatures above 200°C. Do not expose to direct sunlight.

10.5. Incompatible materials

Avoid contact with strong acids and strong oxidizing agents.

10.6. Hazardous decomposition products

None under normal conditions of use and storage.

SECTION 11: Toxicological information

11.1. Information on toxicological effects

Acute toxicity: Based on available data, the classification criteria are not met.

Skin corrosion/irritation: Based on available data, the classification criteria are not met.

Serious eye damage/irritation: Based on available data, the classification criteria are not met.

Respiratory or skin sensitization: Based on available data, the classification criteria are not met.
Germ cell mutagenicity: Based on available data, the classification criteria are not met.
Carcinogenicity: Based on available data, the classification criteria are not met.
Reproductive toxicity: Based on available data, the classification criteria are not met.
STOT-single exposure: Based on available data, the classification criteria are not met.
STOT-repeated exposure: Based on available data, the classification criteria are not met.
Aspiration hazard: Based on available data, the classification criteria are not met.

Inhalation: Dust may cause respiratory tract irritation.
Skin contact: May cause skin irritation.
Eye contact: In high concentration product cause serious conjunctivitis and inflammation of the cornea of the eye.
Ingestion: May cause mucous membranes of mouth and throat irritation.

SECTION 12: Ecological information

12.1. Toxicity
Does not meet the criteria of classification as dangerous for the environment.

12.2. Persistence and degradability
Not specified.

12.3. Bioaccumulative potential
Not specified.

12.4. Mobility in soil
Insoluble in water, not mobile in soil.

12.5. Results of PBT and vPvB assessment
Not applicable.

12.6. Other adverse effects
Not specified.

SECTION 13: Disposal considerations

13.1. Waste treatment methods
During removal of waste comply with the regional/national laws.

Community legislation:
- Directive **2008/98/EC** of the European Parliament and of the Council of 19 November 2008 on waste and repealing certain Directives as amended.
- European Parliament and Council Directive **94/62/EC** of 20 December 1994 on packaging and packaging waste as amended.

Disposal methods for the product: Do not introduce into the environment. Disposal in accordance with the local/national legislation.
Disposal methods for used packing: Empty containers give for appropriate rubbish dump or for disposal in accordance with the local/national legislation.

SECTION 14: Transport information

14.1. UN number
Not applicable.

14.2. UN proper shipping name
Not applicable.

14.3. Transport hazard class(es)
Not applicable.

14.4. Packing group
Not applicable.
14.5. Environmental hazards
Does not meet the criteria of classification as dangerous for the environment.
14.6. Special precautions for user
During cargo handling use personal protective equipment - see Section 8. Do not allow the release of dust during transportation; use airtight containers. Wet activated carbon absorb oxygen from air, and there is possibility of dangerously low oxygen levels.
14.7 Transport in bulk according to Annex II of Marpol and the IBC Code
Not applicable.

SECTION 15: Regulatory information
15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture
Regulation (EC) No 1907/2006 of the European Parliament and of the Council of 18 December 2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH), establishing a European Chemicals Agency, amending Directive 1999/45/EC and repealing Council Regulation (EEC) No 793/93 and Commission Regulation (EC) No 1488/94 as well as Council Directive 76/769/EEC and Commission Directives 91/155/EEC, 93/67/EEC, 93/105/EC and 2000/21/EC as amended.
Regulation (EC) No 1272/2008 of the European Parliament and of the Council of 16 December 2008 on classification, labelling and packaging of substances and mixtures, amending and repealing Directives 67/548/EEC and 1999/45/EC, and amending Regulation (EC) No 1907/2006 as amended.
Commission Regulation (EU) 2015/830 of 28 May 2015 amending Regulation (EC) No 1907/2006 of the European Parliament and of the Council on the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH).
Directive 2008/98/EC of the European Parliament and of the Council of 19 November 2008 on waste and repealing certain Directives as amended.
European Parliament and Council Directive 94/62/EC of 20 December 1994 on packaging and packaging waste as amended.

15.2. Chemical safety assessment
The Chemical Safety Assessment has been performed.

SECTION 16: Other information
The full text of H-statements from Sections 2 and 3:
None.

Key to abbreviations and acronyms:
BLV - Biological Limit Value.
STEL - Short-term exposure limit.
TWA - 8 hours' time-weighted average.

Sources of key data:
Producers SDS from 2nd November 2015 (version 1).

Training advice: Before use read the SDS.

The information above is based on a current available data concerning the product, but also on the experience and knowledge in this field of the producer. They are neither

a quality description of the product nor a guarantee of particular features. They are also treated as aid to safety in transport, storage and usage of the product. This does not free the user from the responsibility of improper usage of the information above also of improper compliance with the law norms in the field.

Prepared by ISOTOP Consulting Company; **www.isotop.pl**; e-mail: **reach@isotop.pl**

SDS from 22nd April 2016 (Version 1) has been revised in sections 1.1, 3.2, 8.1, 15.2. Changes have been underlined.

This SDS replaces and annuls all the previous versions.